

Figure 12.6 X Ray machine



'X rays were discovered by German physicist Sir Wilhelm Conard Roentgen in 1895, while he was studying on high voltage discharge phenomenon in a Crooker's tube. He noticed fluorescence of a barium platino cyanide screen lying several feet away in the same room.

Ultrasound imaging

Ultrasound literally means sound beyond the range of human hearing. Ultrasound waves are produced by a physical phenomenon known as Piezo-electric effect. When an electric potential is applied to certain crystals for example: Lead zirconate, they become excited, vibrate and produce ultrasound. When the ultrasound waves are introduced through homogenous tissue, they pass unimpeded until they meet another tissue or organ. A part or whole of the ultrasound wave is reflected and received back by the same crystal and is converted into an electrical signal. This signal denoting reflecting interface is shown on the oscilloscope screen as a deflection from the base line (Figure12.7).

Clinical significance

1. Ultrasound waves are used to image the foetus at different stages of pregnancy to study the progress of the developing foetus.
2. They are used to hear foetal heart sound, blood flow, etc.
3. Used in echocardiography to diagnose the damages in heart.
4. Used for diagnosis of tumours, gall stones, kidney stones, obstructions in the genital tracts.

Computed Tomographic (CT) Scanning

Computed tomography is originally known as computed axial tomography (CAT or CT Scan). The word tomography



Figure 12.7 Ultrasonography

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